

DEVA ANAND

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Education

MS, Computer Science , University of Massachusetts Amherst	2025–2027
<i>Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering</i>	
BE, Computer Science and Engineering , Vels Institute of Science, Technology and Advanced Studies	2019–2023

Professional Experience

Wysa , Backend Engineer <i>Bengaluru, India</i>	Jul 2023 – Jul 2025
- Shipped features end to end across a multi-tenant NHS eTriage backend (Node.js, MongoDB) for 8+ UK clients, processing 10,000+ monthly submissions via Mayden iaptus REST integrations; contributed to \$340K+ in revenue. - Built a full-stack annotation platform (React front-end, Node.js/MongoDB back-end) used daily by the AI team, cutting manual work by 100+ hours/month and lifting labeling throughput by 60%. - Debugged production incidents across frontend, backend, and integrations to root cause; implemented AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating 20+ VAPT findings. - Served as primary technical POC with high ownership, onboarding a second developer to on-call readiness and driving a COO-sponsored GitHub Copilot AI-assisted-development experiment through PR review.	
Cario Growth Services , Backend Developer Intern <i>Chennai, India</i>	Nov 2022 – May 2023
- Designed and shipped 20+ RESTful APIs (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team. - Migrated core repositories to TypeScript and shipped a production LLM workflow (auto-commenting via 4-bit quantized Llama-7B, llama.cpp) with per-persona prompts, async queued generation, and a content-safety filter, built end to end with no ML mentor on the team.	
Freelance / Open Source , Full-Stack Engineer <i>Remote</i>	Jul 2022 – Aug 2023
- Built and deployed Underdogs Fitness (Website), a production MERN platform (React front-end) with Stripe hybrid payments, cron automation, and multi-branch support; live 3+ years. - Delivered a second MERN client engagement (US-based inventory management system) end-to-end as a paid freelance build.	

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub Demo)	Python, FastAPI, OpenAI / Groq, SQLite
Built a multi-stage agent that senses open-ended queries and acts, planning typed retrieval, reranking pages, extracting entities deterministically with LLM fallback, verifying outputs, and surfacing cell-level provenance through a reviewer-facing trust UI; hardened with 150+ tests and provider routing across OpenAI/Groq.	
cf-ai-research-scout – Edge-Native AI Assistant (GitHub)	TypeScript, Cloudflare Workers AI, Durable Objects, D1
Built an edge-native AI agent in TypeScript that fetches sources, decides multi-step actions, and streams results over WebSocket, using Workers AI for embeddings and inference with durable workflows and D1-backed state that survive process restarts and individual tool failures.	
Sonare – Offline Sign / Speech App (GitHub)	React, Electron, FastAPI, MediaPipe, whisper.cpp
Engineered a React/Electron frontend and FastAPI backend integrating MediaPipe hand tracking, whisper.cpp ASR, and local TTS into one offline pipeline, designing async APIs and queues to keep the UI responsive under low-latency local video and audio streams.	

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, HTML/CSS, PostgreSQL, MongoDB, SQLite
Web & Full-Stack: React, Tailwind CSS, Node.js, Express, FastAPI, REST APIs, Electron
LLM & Agents: LLM agents, agentic pipelines, RAG, OpenAI / Groq APIs, output verification, evaluation
AI / ML: PyTorch, Hugging Face Transformers, sentence-transformers, pipeline instrumentation, cell-level provenance
Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.